

Key: Seventy-Five Seats

84-355 Democracy's Data | Regional shift 1960–2020 | Instructor copy — do not distribute

Jonathan Cervas

August 11, 2026

All code is given to students. Grade the interpretation.

Structure. The brief (`regional-shift-brief.Rmd`) is now a chapter rather than a parallel copy of the lab: it is organised by argument, under idea headings, with four numbered figures and no numbered steps. The lab (`regional-shift.Rmd`) still runs Steps 1–11 with the code exposed. **Students should read the brief first. Step numbers below refer to the lab;** the chapter is cited by figure number or section title.

This session is built around a single reveal, and the chapter now protects it too. The brief holds the total back for several pages and stops the reader with its own prediction box — write down two numbers, New York's total loss and the seats that changed states altogether — immediately before *What the maintenance adds up to*, which is where it reveals 75. A student who read the chapter properly has therefore already guessed, while the guess was still worth something. The lab repeats the structure: Steps 1–4 are deliberately boring — they show one reapportionment, it moves 7 seats, and the room concludes that reapportionment is administrative housekeeping — and Step 5 adds six of them together. **Get the prediction out of the room before Step 5 runs:** ask for the number they wrote down while reading, and collect a fresh guess from anyone who did not read. Everything else in the hour depends on that gap.

Every number below was recomputed from the built files at knit time.

The session in one sentence

Between 1960 and 2020 the United States moved 75 House seats — 17.2% of the chamber, and with them 66 electoral votes — from the Northeast and Midwest to the South and West, without a single vote being taken on any of it, and no individual decade ever looked like more than paperwork.

The findings

```
c(seats_that_changed_states = gross,
  pct_of_the_house          = round(100 * gross / HOUSE, 1),
  crossed_a_regional_border = crossed,
  largest_single_decade     = max(byd),
  smallest_single_decade    = min(byd),
  new_york_loss             = abs(ny$change),
```

```

states_smaller_than_that = sum(sc$reps_2020 <= abs(ny$change)),
ne_mw_seats_1960         = b60[["NE+MW"]],
ne_mw_seats_2020         = b20[["NE+MW"]],
ne_mw_pct_2020           = round(100 * b20[["NE+MW"]] / HOUSE, 1),
ne_mw_ev_pct_1960        = round(100 * ev60[["NE+MW"]] / sum(ev60), 1),
ne_mw_ev_pct_2020        = round(100 * ev20[["NE+MW"]] / sum(ev20), 1),
one_way_states           = sum(mono),
people_per_seat_multiple = round(P2 / P1, 2),
population_multiple       = round(nat$pop[nat$year == Y2] /
                                nat$pop[nat$year == 1910], 2))

```

```

#> seats_that_changed_states      pct_of_the_house crossed_a_regional_border
#>                75.00                17.20                66.00
#>   largest_single_decade  smallest_single_decade      new_york_loss
#>                19.00                7.00                15.00
#>   states_smaller_than_that      ne_mw_seats_1960      ne_mw_seats_2020
#>                44.00                233.00                167.00
#>          ne_mw_pct_2020      ne_mw_ev_pct_1960      ne_mw_ev_pct_2020
#>                38.40                51.40                39.10
#>          one_way_states  people_per_seat_multiple      population_multiple
#>                47.00                3.62                3.59

```

- **75 seats changed states, 66 of them left their region entirely.** The shift is not neighbours trading; it is one half of the country moving to the other.
- **No decade did it.** The largest reapportionment in the window moved 19 seats, the smallest 7. The finding is entirely a product of addition.
- **New York lost 15**, and 44 of the 50 states do not have 15 seats in total.
- **The Northeast and Midwest went from a House majority (233 of 435) to 38.4%**, and from 51.4% of the fifty states' electoral votes to 39.1%.
- **47 of 50 states moved in one direction only.** A ratchet, not a cycle.
- **A representative now speaks for 3.62× as many people as in 1910**, and the country grew 3.59×. The near-identity is the mechanism: the denominator was frozen in 1929.

One arithmetic detail worth having ready, because a sharp student will ask why the per-decade gains do not sum to the cumulative total:

```

c(sum_of_per_decade_gains = sumdec,
  cumulative_1960_to_2020 = gross,
  difference                = sumdec - gross,
  states_that_reversed     = length(NONMONO))

```

```

#> sum_of_per_decade_gains  cumulative_1960_to_2020      difference
#>                78                75                3
#>   states_that_reversed
#>                3

```

NONMONO

```

#> [1] "California" "Montana"      "Tennessee"

```

The gap is exactly the number of states that moved in both directions (California, Montana, Tennessee), each of which gained a seat it later gave back or gave one back it later regained. **Gross movement exceeds**

net movement by precisely 3, in sixty years. That is the ratchet stated as an identity rather than as a metaphor.

The three things to get said

- 1. The unit of observation is doing the work.** The same file, honestly reported, supports “reapportionment is minor administrative housekeeping” and “reapportionment moved a sixth of the House.” Neither is a lie. The difference is whether you look at one decade or six. **Make somebody say why a news cycle produces the first framing** — because reporting is organised around events, and this is not an event. It is a rate. The course’s recurring question, *what window is your claim about?*, has no cleaner example.
- 2. Nobody decided this, and that is the interesting part.** Students arrive expecting political change to have an author. Ask who is responsible for the transfer of 66 electoral votes between regions. There is no answer: no bill, no vote, no litigant, no agency. The Constitution specifies a procedure and the procedure ran. **A student who says “so it is nobody’s fault” has half of it; push to the other half — the rules that convert population into power were chosen, and choosing not to revisit them is also a choice.** The House has been 435 since 1929.
- 3. Step 8 is the methodological spine, and it is the step that is easiest to skip.** “The South” has no agreed definition, the lab computes the finding under several, and the answer moves — but never reverses. That combination is what an honest sensitivity analysis looks like, and students almost never see one. See the dedicated section below; **do not let this become a two-minute aside.**
- The line for the board:** power moved without anybody moving it, and the only reason it looks like nothing happened is that we keep checking every ten years.

Walking the steps

Step	What it does	Min	Watch for
1–3 One record, the file, why 1960	Alaska and Hawaii have no seats before 1960	7	Do not skip the de-comma trap. "4,951,560" becomes NA silently. It is the cheapest data-literacy lesson in the course.
4 One decade at a time	7 seats, 13 states in 2020	8	Let them conclude it is trivial. The step is supposed to underwhelm.
— The prediction	Written down, before Step 5	4	Non-negotiable. Collect guesses aloud. Typical answers are 20–30; the answer is 75.

Step	What it does	Min	Watch for
5 The pivot	75 seats, 17.2% of the House	10	The session. Sit in the silence. Then the New York comparison: 44 of 50 states are smaller than what one state lost. The monotonicity check is what upgrades “a shift” to “a trend”. Students read it as a technicality. The South’s rise begins after 1960, not before. Between 1910 and 1960 its share <i>fell</i> . Nearly everyone assumes otherwise. The methods step. See below. Take a vote on which definition is “right” before showing the table. Separate <i>bigger</i> districts from <i>less equal</i> districts — the ratio actually improved (2.14→1.83). Keep it arithmetic. The file has no election results; do not let the room infer a partisan story it cannot support. The “why” is not in this file. Name labs/migration out loud.
6 Regions and the ratchet	233 → 167; 47 of 50 states one-way	8	
7 The long line	Population share, 1910–2020	8	
8 Defining the South	Four definitions, one direction	10	
9 What a seat is worth	210,328 → 761,169 people per seat	6	
10 Electoral votes	51.4% → 39.1%	6	
11 What you can say	Including what you cannot	5	

If you are short on time, protect the prediction, Step 5 and Step 8. The underwhelming decade, the sixty-year total, and the demonstration that the total does not depend on how you drew the category are the whole argument. Steps 9 and 10 are extensions.

The South-definition sensitivity, and how to use it

This is the most transferable thing in the lab and it needs ten minutes, not two. The lab hands students the finding “the South rose”; Step 8 asks whether that sentence survives a change in who counts as Southern.

Run the vote first. Ask the room: *is Maryland Southern? Oklahoma? Delaware? West Virginia? the District of Columbia?* You will get disagreement in under a minute, and the disagreement is the point — these are not edge cases invented for a seminar. The Census Bureau counts all five as South. **house-competition** and **historical-campaigns**, both in this course, count none of them.

Then show what it does to the answer.

```

defs <- list(
  "11 Confederate states"      = CONFED,
  " + Maryland"               = c(CONFED, "Maryland"),
  " + Oklahoma"               = c(CONFED, "Oklahoma"),
  " + Kentucky, West Virginia" = c(CONFED, "Kentucky", "West Virginia"),
  " + District of Columbia"    = c(CONFED, "District of Columbia"),
  "Census South (16 + DC)"     = c(CONFED, BORDER),
  "Sun Belt (South + West)"    = c(CONFED, BORDER,
    unique(st$name[st$region == "West"])))

out <- do.call(rbind, lapply(names(defs), function(k) {
  s <- share_of(defs[[k]])
  data.frame(definition = k, states = length(defs[[k]]),
    share_1910 = round(s[["1910"]], 2),
    share_1960 = round(s[[as.character(Y1)]], 2),
    share_2020 = round(s[[as.character(Y2)]], 2),
    rise_1960_2020 = round(s[[as.character(Y2)]] - s[[as.character(Y1)]], 2),
    drift_1910_1960 = round(s[[as.character(Y1)]] - s[["1910"]], 2))
}))
out

```

Definition	States	Share 1910	Share 1960	Share 2020	Rise 1960 2020	Drift 1910 1960
11 Confederate states	11	24.28	24.22	32.63	8.41	-0.06
+ Maryland	12	25.68	25.95	34.49	8.54	0.27
+ Oklahoma	12	26.08	25.52	33.82	8.30	-0.56
+ Kentucky, West Virginia	13	28.09	26.95	34.53	7.58	-1.13
+ District of Columbia	12	24.64	24.65	32.84	8.19	0.01
Census South (16 + DC)	17	31.87	30.66	38.10	7.44	-1.21
Sun Belt (South + West)	30	39.54	46.30	61.81	15.51	6.76

Four things to draw out of that table.

The direction never changes. Every definition rises, and rises by at least 7.4 points. **A finding that survives every plausible operationalisation of its central category is a robust finding**, and this is the sentence to make them say. It is not the same as a finding that is big — it is a finding that is not an artefact of a coding decision.

The magnitude does change, and by a defensible amount. From 7.44 points to 8.54 across the South definitions — a spread of 1.10 points, or 12.9% of the largest estimate. **So a student who reports the number to two decimals without saying which South they used has overstated their precision**, even though they have not overstated their conclusion.

The narrow definition gives the larger answer. This is counter-intuitive and worth pausing on: adding states to a growing category makes the growth look *smaller*. It can only mean the added states are a drag, and they are. The border bloc as a whole went from 6.4% of the country to 5.5%.

But not all of them — and this is the reward for a student who checks. Two of the six additions make the rise bigger, and Maryland is much the larger of the two:

```

b <- do.call(rbind, lapply(BORDER, function(v) {
  s <- share_of(v)
  data.frame(state = v, share_1960 = round(s[[as.character(Y1)]], 3),
             share_2020 = round(s[[as.character(Y2)]], 3),
             change = round(s[[as.character(Y2)]] - s[[as.character(Y1)]], 3))
}))
b[order(-b$change), ]

```

State	Share 1960	Share 2020	Change
Maryland	1.729	1.864	0.135
Delaware	0.249	0.299	0.050
Oklahoma	1.298	1.195	-0.104
District of Columbia	0.426	0.208	-0.218
Kentucky	1.694	1.359	-0.335
West Virginia	1.037	0.541	-0.496

Maryland and Delaware grew slightly faster than the country; Oklahoma, West Virginia, Kentucky and District of Columbia did not. **The Census South is a mixture of two demographic regimes wearing one label**, which is a better criticism of the category than “it includes Maryland and Maryland is not really Southern.” The first is measurable; the second is an argument about identity that this file cannot adjudicate.

Where to land it. The Census Bureau’s regions are not wrong — they are a statistical convenience with a legal and historical basis, and the Bureau publishes its own totals so you can check your mapping against them, which `data/build-data.R` does. The lesson is that **a category boundary is a hypothesis, and a finding is only as strong as its behaviour when you move the boundary**. Students should leave able to ask, of any regional or demographic claim: *who is in the category, who decided, and does the answer move?*

Option A — Move the window

Verified output

```

a <- do.call(rbind, lapply(WSTART, function(S) {
  w <- win(S)
  data.frame(window = paste(S, "to", Y2), decades = (Y2 - S) / 10,
             seats_moved = sum(w$ch[w$ch > 0]),
             per_decade = round(sum(w$ch[w$ch > 0]) / ((Y2 - S) / 10), 2),
             NE_MW = sum(w$ch[w$region %in% c("Northeast", "Midwest")]),
             S_W = sum(w$ch[w$region %in% c("South", "West")]))
}))
a$pct_of_1960_total <- round(100 * a$seats_moved / a$seats_moved[1], 1)
a

```

Window	Decades	Seats moved	Per decade	NE MW	S W	Pct of 1960 total
1960 to 2020	6	75	12.5	-66	66	100.0
1970 to 2020	5	65	13.0	-58	58	86.7
1980 to 2020	4	48	12.0	-41	41	64.0
1990 to 2020	3	30	10.0	-26	26	40.0
2000 to 2020	2	19	9.5	-16	16	25.3
2010 to 2020	1	7	7.0	-5	5	9.3

```
half <- a[a$window == paste(1990, "to", Y2), ]
c(full_window_seats = a$seats_moved[1],
  half_window_seats = half$seats_moved,
  half_of_full      = a$seats_moved[1] / 2,
  ratio             = round(half$seats_moved / a$seats_moved[1], 3))
```

```
#> full_window_seats half_window_seats      half_of_full      ratio
#>                75.0                30.0                37.5                0.4
```

The lab asks a specific question and it has a specific answer. A window half as long (3 decades instead of 6) yields 30 seats, not the 37.5 a constant rate would give. The rate is **falling**: 12.5 seats per decade over the full window, 7.0 in the most recent one.

3/4: Changes **START**, reports that a shorter window moves fewer seats, and gets the direction right — it is less than half, so the shift is decelerating rather than accelerating.

4/4: Computes the per-decade rate rather than eyeballing the total, and notices that the ranking of gainers and losers barely moves while the magnitudes shrink — Texas, Florida and Arizona lead every window; New York, Ohio, Pennsylvania, Illinois and Michigan trail every window. **The composition is stable; only the speed changes.** That distinction is the answer.

4/4, best: Takes the confidence question seriously, which the prompt asks for and most students skip. With 6 observations of a per-decade rate, “decelerating” is a description of six numbers, not an estimate with a standard error — and the most recent point (7 seats in 2020) is the smallest in the series and therefore the most tempting and least reliable place to draw a trend. A student who says *this could be a slowdown, or it could be that the 2020 census undercounted, or it could be one decade of noise, and this file cannot separate them* has understood what six observations buy you. **Reward explicitly.**

Option B — Redefine the South

Verified output

The full sweep is in the sensitivity section above. Students will typically run three or four variants; these are the ones the prompt names.

```
out[, c("definition", "states", "share_1960", "share_2020", "rise_1960_2020")]
```

Definition	States	Share 1960	Share 2020	Rise 1960 2020
11 Confederate states	11	24.22	32.63	8.41
+ Maryland	12	25.95	34.49	8.54
+ Oklahoma	12	25.52	33.82	8.30
+ Kentucky, West Virginia	13	26.95	34.53	7.58
+ District of Columbia	12	24.65	32.84	8.19
Census South (16 + DC)	17	30.66	38.10	7.44
Sun Belt (South + West)	30	46.30	61.81	15.51

3/4: Edits MY_SOUTH, reports that the share changes, and notices the rise happens regardless.

4/4: Identifies *which* additions shrink the rise and says why — the added states grew slower than the country, so they dilute a growing category. Connects that back to the finding: the Census definition **understates** the shift, so using it is the conservative choice.

4/4, best: Answers the harder half of the prompt — that the variable encodes an argument. The eleven-state list defines the South by a political event of the 1860s; the Census list defines it by a line on a map maintained for statistical convenience. **These are claims about what a region *is*, and they are not interchangeable even though they produce similar answers here.** The strongest version notes that the two happen to agree in this application and would not agree in others — a lab about Reconstruction-era voting rights and a lab about Sun Belt housing markets should not use the same South — and then says which one this lab should use and why. Any defensible choice earns the mark; **the reasoning is the assignment.**

The Maryland exception (0.135 points, the larger of the two border additions that make the rise *larger*) is now priced out in the chapter itself, under *Whether the South is eleven states or seventeen*, so naming it no longer earns anything on its own. **Reward instead the student who uses it:** who sees that the six border units are a drag in aggregate while two of them are not, and concludes that “the border states behave like the Northeast” is a statement about the aggregate that at least one member of the aggregate contradicts. That is the same reading skill the sensitivity table teaches, applied one level down.

Option C — Growth is not seats

Verified output

```
c(correlation_all_states = round(cor(sc$pop_growth, sc$change), 2),
  national_growth_pct    = round(NG, 1),
  fastest_growing_state  = round(max(sc$pop_growth)),
  its_seat_gain          = sc$change[which.max(sc$pop_growth)])
```

```
#> correlation_all_states    national_growth_pct    fastest_growing_state
#>                0.45                84.80                988.00
#>            its_seat_gain
#>                3.00
```

```
big <- sc[sc$reps_1960 >= 10, ]
c(states_with_10plus_seats_in_1960 = nrow(big),
  correlation_among_those          = round(cor(big$pop_growth, big$change), 2),
```



```
correlation_with_size_x_relative_growth =
  round(cor(sc$change, sc$reps_1960 * (sc$pop_growth - NG)), 2))
```

```
#>          states_with_10plus_seats_in_1960          correlation_among_those
#>                                     16.00                                0.89
#> correlation_with_size_x_relative_growth
#>                                     0.99
```

```
x <- sc[sc$pop_growth > NG & sc$change <= 0,
       c("name", "region", "pop_growth", "reps_1960", "change")]
x$pop_growth <- round(x$pop_growth, 1)
x[order(x$reps_1960), ]
```

Name	Region	Pop growth	Reps 1960	Change
Alaska	West	224.3	1	0
Delaware	South	121.8	1	0
Hawaii	West	130.0	2	0
Idaho	West	175.6	2	0
New Hampshire	Northeast	127.0	2	0
Maryland	South	99.2	8	0
Tennessee	South	93.7	9	0

```
c(median_1960_delegation_of_those = median(x$reps_1960),
  median_1960_delegation_all      = median(sc$reps_1960))
```

```
#> median_1960_delegation_of_those    median_1960_delegation_all
#>                                2                                6
```

The two reasons the prompt asks for, both visible in the scatter:

1. **Percentage growth ignores base size.** Nevada grew 988% from a delegation of 1; Florida grew 335% from a delegation of 12, and gained 16 seats to Nevada's 3. Seats follow *people*, and a large percentage of a small number is a small number.
2. **Apportionment is relative, not absolute.** The national benchmark is 84.8% growth. Growing is not enough; you have to grow faster than everyone else, and by enough to cross a whole-seat threshold.

Put those together and the correlation stops being disappointing: 0.99 between seat change and *1960 delegation size* $\times$ *growth above the national rate*. **The raw 0.45 is not weak measurement; it is the wrong quantity.**

3/4: Reports the correlation, names Nevada and Florida, and gets at least one of the two reasons.

4/4: Gets both reasons and identifies what the 7 no-gain fast-growers have in common — they are small (median 2 seats in 1960 against 6 for all states) or they beat the national rate only narrowly (Tennessee and Maryland). Whole seats are a coarse instrument, and a state with one seat cannot be given nine tenths of another.

4/4, best: Reframes the correlation as a measurement question rather than a finding — asks what quantity *would* correlate with seat change, and constructs it, or gets close in words. A student who says *the right predictor is share of national population, not growth in own population* has stated the apportionment rule from scratch, which is the whole content of apportionment arrived at independently. **This is the strongest response the lab can produce; mark it as such.**

Option D — Ask your own question

Ungraded on content, graded on whether the question is answerable with this file. The 1920s prompt is the one most students take, so here is what it contains:

```
nat[nat$year %in% c(1910, 1920, 1930), c("year", "pop", "reps", "pop_per_rep")]
```

Year	Pop	Reps	Pop per rep
1910	92228531	433	210328
1920	106021568	435	241864
1930	123202660	435	280675

```
chg <- st[st$year == 1920 & !is.na(st$repchg) & st$repchg != 0,  
        c("name", "year", "reps", "repchg")]  
chg
```

Name	Year	Reps	Repchg
Arizona	1920	1	1
New Mexico	1920	1	1

```
c(states_with_any_change_in_1920 = nrow(chg),  
   states_in_1970_for_comparison = sum(st$year == 1970 & !is.na(st$repchg) &  
                                         st$repchg != 0))
```

```
#> states_with_any_change_in_1920 states_in_1970_for_comparison  
#>                                2                             14
```

Congress refused to reapportion after the 1920 census — the only time it has happened — and the file records the refusal as an absence. Every state's change is zero except Arizona and New Mexico, which are not reapportionment at all: they became states in 1912 and enter the seat columns for the first time. **The absence of an event is the observation**, which is a genuinely hard thing to see in a data file and worth pointing out to anyone who takes this option.

The population columns show the country that produced the refusal: the 1920 census was the first to find a majority of Americans living in urban areas, and reapportioning would have moved seats from rural states to cities. The file shows the population change and is completely silent on the motive. **A student who asserts the motive from this file has made exactly the error the lab is about;** a student who says *the file shows the seats did not move and cannot tell me why* has got it right.

Two smaller quirks in the source, in case they surface:

- **Change in Number of Representatives** is the Bureau's own column and disagrees with the difference of **Number of Representatives** in exactly two rows, both Arizona and New Mexico in 1930, both outside the 1960–2020 seat window.
- 1910 apportioned 433 seats, not 435; Arizona and New Mexico brought it to 435 after statehood, where the Permanent Apportionment Act of 1929 froze it.

Grading

Four-point scale from the syllabus. Two or three paragraphs, one chosen chunk.

- **Never penalise code.** It is given, and several options require changing one literal — a student who changed it correctly has done everything asked.
- **The move that earns a 4** is treating a methodological choice — the window, the category boundary, the quantity correlated — as a substantive decision with consequences, and saying what those consequences are.
- **A 2** is “the South grew and the Northeast shrank.” True, in the brief, and requires nothing of the student.
- **Also a 2:** any claim about *why* people moved. The file has population and seats. Jobs, weather, housing and air conditioning are not in it, and the brief says so twice.
- **Also a 2:** treating 75 as large or small without a comparison. The number means nothing until it is put next to something — the size of a delegation, a decade’s movement, the margin in the House.
- **Reward the reversal.** A student who notices that the same data supports two opposite-sounding framings, and explains what makes each one honest, has the point of the session.
- **Reward anyone who asks for uncertainty.** There is none in this file: populations print as exact integers, coverage error is published elsewhere. Noticing its absence is the most sophisticated available response, and it is the same instinct **apportionment** rewards for a case where an 89-person margin decided a seat.

Anticipated questions

- “*Why not go back to 1910? The population data is there.*” — Because the seat data would be comparing different countries. Alaska and Hawaii are in the file from 1910 with populations and no representatives. The lab uses 1910 population where it is honest to (Step 7, labelled) and 1960 seats everywhere else. **This question comes up most sessions and the answer is a good one to have crisp.**
- “*Isn’t this just California and Texas?*” — No, and the bar chart answers it: 35 states changed size and 21 lost seats. But the follow-up is real — the top four gainers account for 51 of the 75 seats. Concentration and breadth are both true.
- “*Does the South include Texas?*” — In every definition used here, yes. Texas is 15 seats of the South’s 31 net gain, so a student who wants to argue Texas is its own thing is making a real point. Let them test it: remove Texas from MY_SOUTH in Option B and see what is left.
- “*Do the electoral-vote numbers mean Republicans gained?*” — Not from this file. The regions that gained have voted differently in different decades, and there is no election result anywhere in the data. **This is worth being firm about;** it is the single most likely place for a student to overreach.
- “*Will it keep going?*” — 7 seats moved in 2020, the smallest reapportionment in the window. Either a slowdown or noise. Twelve observations cannot tell you, and Option A is where a student can see that for themselves.
- “*Why is the House still 435?*” — Statute, not the Constitution; Permanent Apportionment Act of 1929. **apportionment** runs that argument properly and this lab deliberately does not.

Connections

- **apportionment** — the same machinery over one decade instead of six. That lab takes a single reapportionment apart, down to New York missing the 435th seat by 89 people; this one never opens the formula and asks only what sixty years of it add up to. **Run apportionment first.** The arithmetic is assumed here, not taught.
- **migration** — the mechanism this lab can measure the consequence of and cannot explain. Every “why” question raised in this session belongs there, and saying so out loud is what keeps Step 11 honest.

- **house-competition** and **historical-campaigns** — both use the eleven-state Confederate South. Step 8 is the place to say that the course itself uses two different Souths and that this is a defensible choice rather than an inconsistency, provided each is stated.
- **precinct-geography** — the other lab where two defensible operational choices give different numbers from the same data. There it was areal versus population weighting; here it is which states are Southern. Same lesson, and students who have done both should be asked to name it.